

Solution Requirements

Date	05 July 2026
Team ID	
Project Name	Personalized Networking Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

The Personalized Networking Assistant uses Artificial Intelligence, Natural Language Processing, DistilBERT, GPT-2, and Wikipedia API to help users build meaningful professional connections. The system analyzes user profiles and event details to generate personalized conversation starters and verify information. The system requirements are categorized into Functional Requirements (FR) and Non-Functional Requirements (NFR).

- **Functional Requirements (FR):** Define the core functionalities of the Personalized Networking Assistant, including user authentication, profile management, AI-based conversation generation, event analysis, fact verification, and conversation history management.
- **Non-Functional Requirements (NFR):** Define the quality attributes of the system, such as performance, security, reliability, usability, scalability, and maintainability.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely register and log in to the application.	High
2	User Profile Management	Users can create and update their profile, interests, and networking goals	High
3	External interfaces	Integrates with Wikipedia API and AI models for fact verification and conversation generation.	High

4	AI Conversation Generation	Generates personalized conversation starters based on user profile and event context.	High
5	Event Theme Analysis	Analyzes event topics using DistilBERT to recommend relevant discussion topics.	High
6	Conversation History	Stores previous conversations and user interactions for future reference.	Medium
7	Feedback & Recommendations	Collects user feedback to improve future AI suggestions.	Medium
8	Cloud Deployment	Supports deployment on IBM Cloud or similar cloud platforms.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Generate AI conversation suggestions quickly.	Response within 2 seconds
2	Scalability	Support multiple users simultaneously.	Cloud deployment supported
3	Security & Data Privacy	Protect user profile and conversation data.	Secure authentication & encrypted storage
4	Reliability & Availability	Ensure continuous system availability.	99% system availability

5	Usability & Accessibility	Provide a simple and intuitive interface.	User-friendly Streamlit web application
6	Other	Support easy updates to AI models and features.	Modular and maintainable architecture